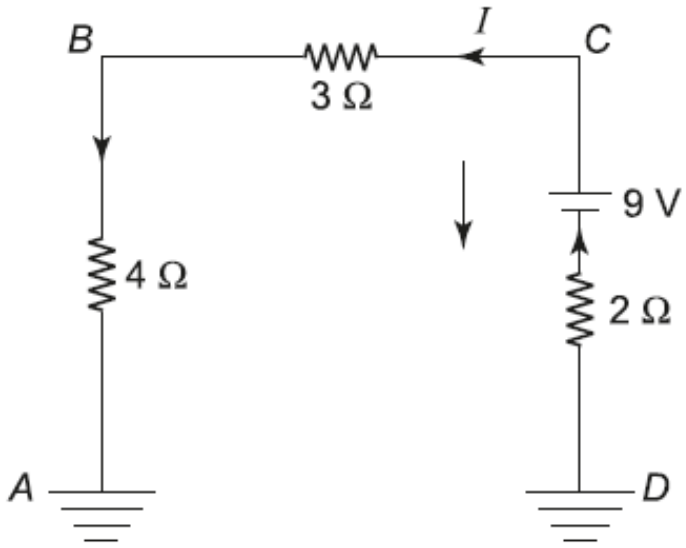
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Find the value of current I in the circuit shown in Fig.



(A) 1 A

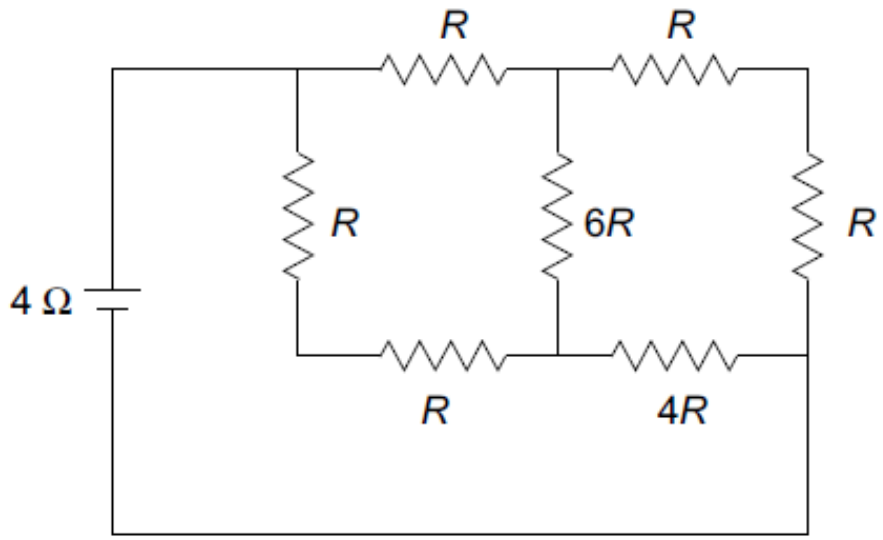
(B) 1.8 A

(C) 1.6 A

(D) 2.4 A

Q.2

A battery of internal resistance 4Ω is connected to the network of resistances as shown in Fig. In order that maximum power can be delivered to the network, the value of R in ohm should be



(A) $\frac{4}{9}$

(B) 2

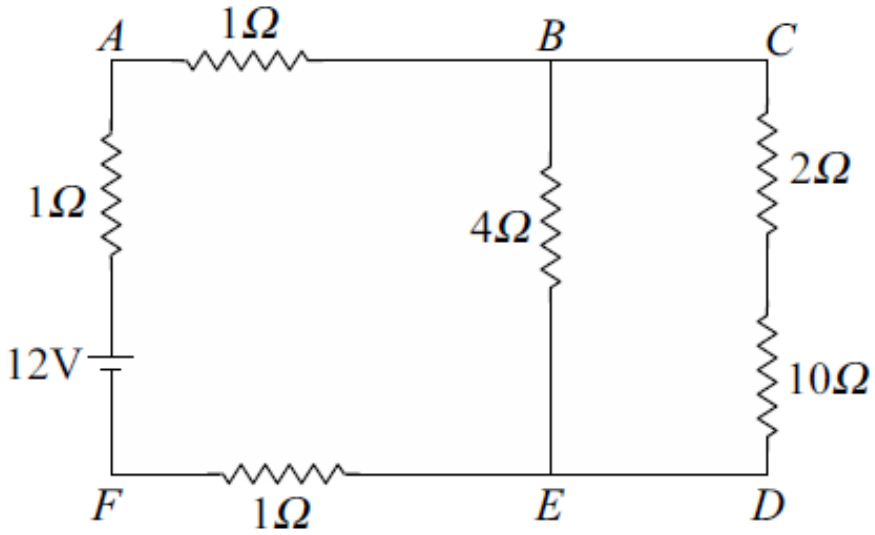
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(C) $\frac{8}{3}$

(D) 18

Q.3

What is the potential difference between points B and D in the circuit shown in Fig. ?



(A) 2 V

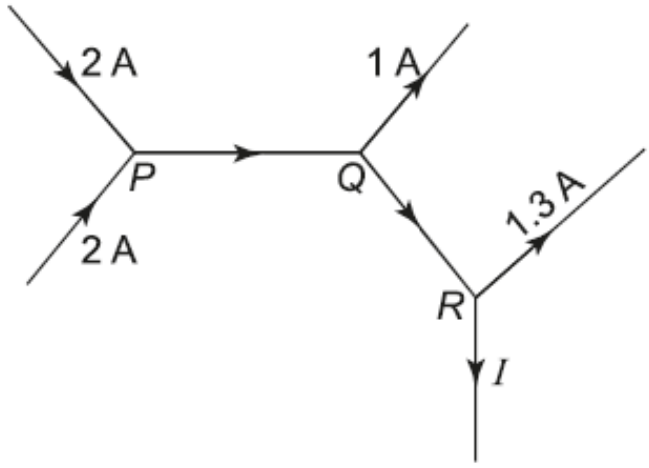
(B) 4 V

(C) 6 V

(D) 8 V

Q.4

Figure shows currents in a part of an electrical circuit. Find the value of current I.



(A) 1.7 A

(B) 1.8 A

(C) 1.6 A

(D) 2.4 A

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Q.5

To get maximum current in a resistance of 3 ohms, one can use n rows of m cells (connected in series) connected in parallel. If the total number of cells is 24 and the internal resistance of a cell is 0.5 ohms then

(A) $\mathrm{m}=12, \mathrm{n}=2$

(B) $m=8, n=3$

(C) $\mathrm{m}=2, \mathrm{n}=12$

(D) $\mathrm{m}=6, \mathrm{n}=4$

Q.6

A car battery of e.m.f. 12 V and internal resistance 5×10^{-2}ohm receives a current of 60 A from an external source, then terminal potential difference of battery is :-

(A) 32 V

(B) 10 V

(C) 15 V

(D) 50 V

Q.7

If n cells each of emf ε and internal resistance are connected in parallel, then the total emf and internal resistances will be

(A) $\varepsilon, r/n$

(B) ε, nr

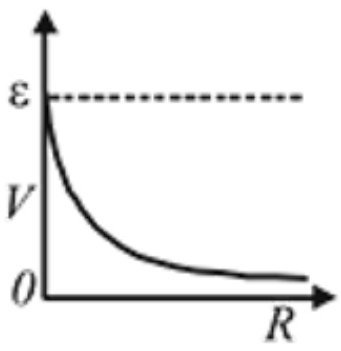
(C) $n\varepsilon, \frac{r}{n}$

(D) $n\varepsilon, nr$

Q.8

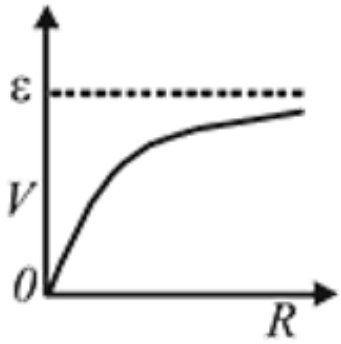
A cell having an emf ε and internal resistance r is connected across a variable external resistance R. As the resistance R is increased, the plot of potential difference V across R is
given by

(A)

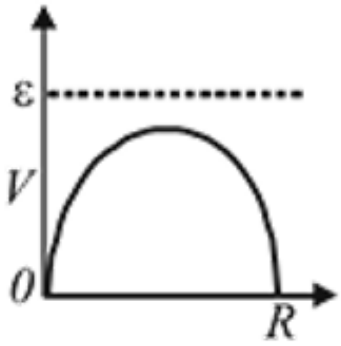


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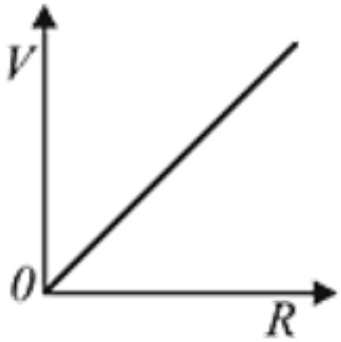
(B)



(C)



(D)



Q.9

The internal resistance of a 2.1 V cell which gives a current of 0.2 A through a resistance of 10Ω is

(A) 0.5Ω

(B) 0.8Ω

(C) 1.0Ω

(D) 0.2Ω

Q.10

A primary cell has an e.m.f. of 1.5 volt. When short-circuited it gives a current of 3 ampere. The internal resistance of the cell is

(A) 4.5ohm

(B) 2ohm

(C) 0.5ohm

(D) $(1/45)$ ohm

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Q.11

In the series combination of n cells each cell having emf ε and internal resistance r. If three cells are wrongly connected, then total emf and internal resistance of this combination will be

(A) $n\varepsilon, (nr - 3r)$

(B) $(n\varepsilon - 2\varepsilon)nr$

(C) $(n\varepsilon - 4\varepsilon), nr$

(D) $(n\varepsilon - 6\varepsilon), nr$

Q.12

A battery of emf $10\ \text{V}$ and internal resistance 3ohm is connected to a resister. The current in the circuit is 0.5 amp. The terminal voltage of the battery when the circuit is closed is

(A) $10\ \text{V}$

(B) zero

(C) $1.5\ \text{V}$

(D) $8.5\ \text{V}$

Q.13

Five cells each of $\text{emf}\ E$ and internal resistance r send the same amount of current through an external resistance R whether the cells are connected in parallel or in series. Then theratio $\left(\frac{R}{r}\right)$ is

(A) 2

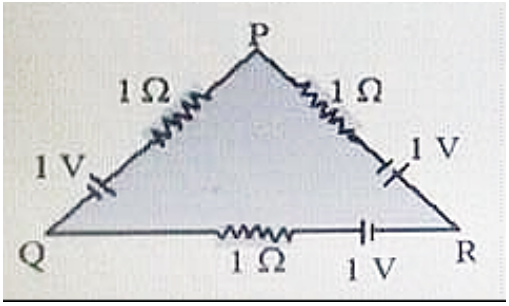
(B) $\frac{1}{2}$

(C) $\frac{1}{5}$

(D) 1

Q.14

Three batteries of emf $1\ \text{V}$ and internal resistance 1Ω each are connected as shown. Effective emf of combination between the points PQ is



(A) zero

(B) $1\ \text{V}$

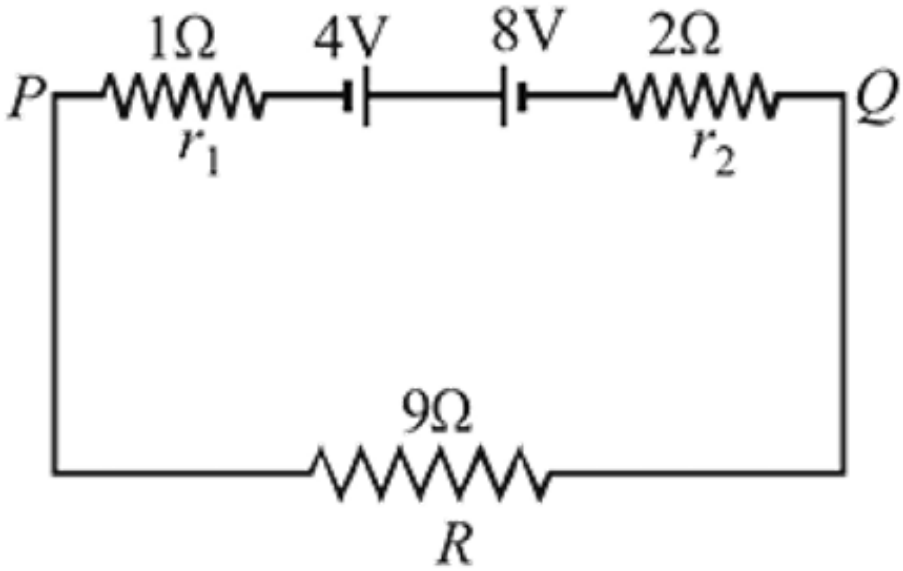
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(C) $2\ \mathrm{V}$

(D) $(2/3)\mathrm{V}$

Q.15

Two batteries of emf $4\ \mathrm{V}$ and $8\ \mathrm{V}$ with internal resistance 1Ω and 2Ω are connected in a circuit with a resistance of 9Ω as shown in figure. The current and potential difference between the points P and Q are



(A) $\frac{1}{3}A$ and 3 V

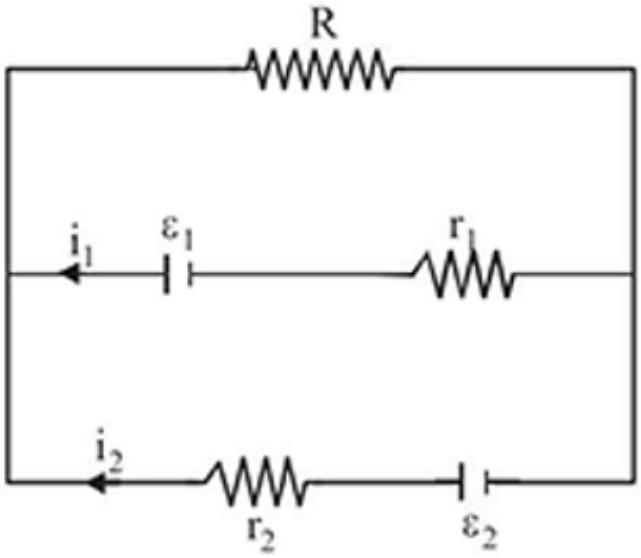
(B) $\frac{1}{6}A$ and 3 V

(C) $\frac{1}{9}A$ and 3 V

(D) $\frac{1}{12}A$ and $12\ \mathrm{V}$

Q.16

See the electric circuit shown in the figure. Which of the following equations is a correct equation for it?



(A) $\varepsilon_2 - \mathrm{i}_2\mathrm{r}_2 - \varepsilon_1 - \mathrm{i}_1\mathrm{r}_1 = 0$

(B) $-\varepsilon_2 - (\mathrm{i}_1 + \mathrm{i}_2)\,\mathrm{R} + \mathrm{i}_2\mathrm{r}_2 = 0$

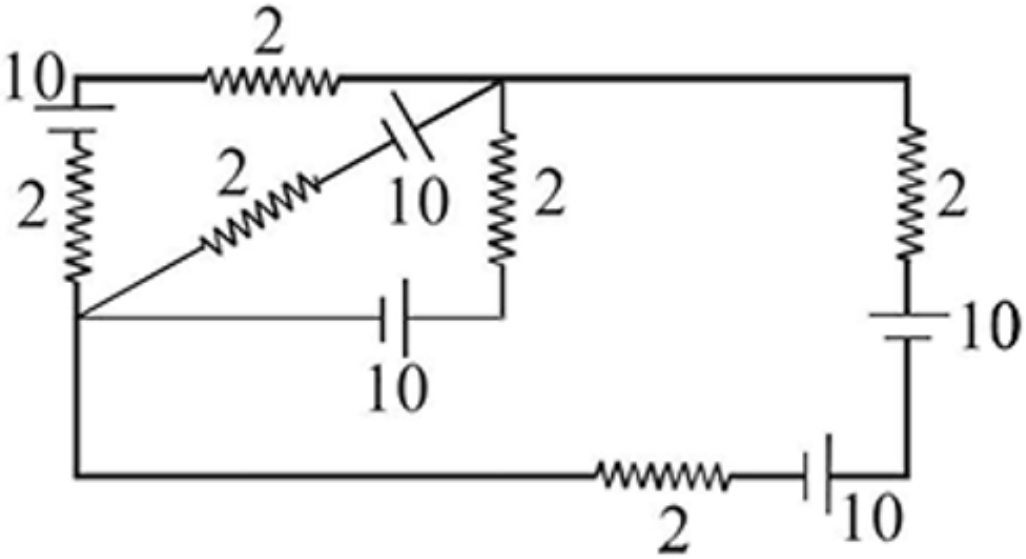
(C) $\varepsilon_1 - (\mathrm{i}_1 + \mathrm{i}_2)\,\mathrm{R} + \mathrm{i}_1\mathrm{r}_1 = 0$

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(D) $\varepsilon_1 - (i_1 + i_2)\,R - i_1 r_1 = 0$

Q.17

All batteries are having emf 10 volt and internal resistance negligible. All resistors are in ohms. Calculate the current in the right most 2Ω resistor.



(A) $\frac{25}{12}$ A

(B) $\frac{25}{6}$ A

(C) $\frac{12}{25}$ A

(D) $\frac{6}{25}$ A

Q.18

The Kirchhoff's second law $(\Sigma iR = \Sigma E)$, where the symbols have their usual meanings, is based on

(A) conservation of momentum

(B) conservation of charge

(C) conservation of potential

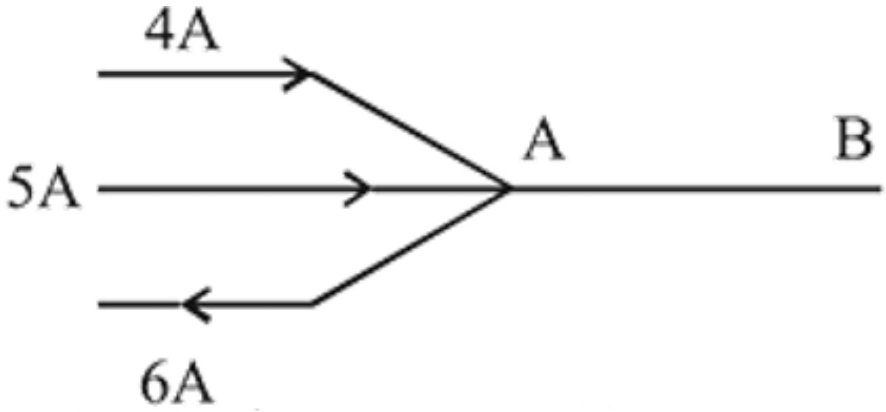
(D) conservation of energy

Q.19

The four wires from a larger circuit intersect at junction A as shown. What is the magnitude and direction of the

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current between points A and B ?



(A) 2 A from A to B

(B) 2 A from B to A

(C) 3 A from A to B

(D) 2 A from B to A

Q.20

A galvanometer coil has a resistance 90 Ω and full scale deflection current 10 mA. A 910 Ω resistance is connected in series with the galvanometer to make a voltmeter. If the least count of the voltmeter is 0.1 V, the number of divisions on its scale is

(A) 50

(B) 100

(C) 120

(D) 80

Answers & Solutions

Q.1 Answer:

1 A

Solution:

$V_A - V_D = -4I - 3I + 9 - 2I$

$= -9I + 9$

But $V_A = V_D$ because points A and D are earthed.

$\therefore \quad -9I + 9 = 0 \Rightarrow I = 1$ A

Q.2 Answer:

2

Solution:

The given circuit is a balanced Wheatstone bridge. [Fig. 1]

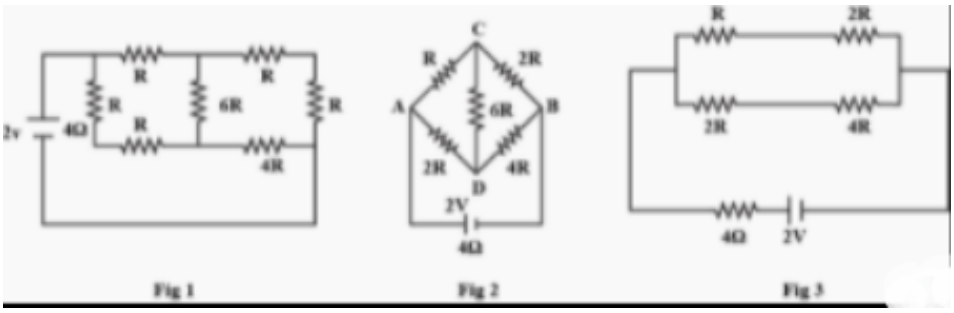
No current will flow across $6\mathrm{R}$ o the side CD. [Fig. 2]

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∴ The equivalent circuit is:- [Fig. 3]

For maximum power net external resistance=Total internal resistance

$$\therefore (3\mathrm{R} \mid 6\mathrm{R})\Omega = 4\Omega$$
$$\implies \frac{3\mathrm{R}\times 6\mathrm{R}}{3\mathrm{R}+6\mathrm{R}}\Omega = 4\Omega$$
$$\implies \mathrm{R} = 2\Omega$$



Q.3 Answer:

6 V

Solution:

Figure shows the assumed directions of currents.

From junction rule, $I = I_1 + I_2$ **Applying the loop rule to loop** $ABEFA$**,**

$$I \times 1 + I_2 \times 4 + I \times 1 - 12 + I \times 1 = 0$$
$$\Rightarrow 3I + 4I_2 = 12$$
$$\Rightarrow 3\left(I_1 + I_2\right) + 4I_2 = 12$$
$$\Rightarrow 3I_1 + 7I_2 = 12 \text{ (1)}$$

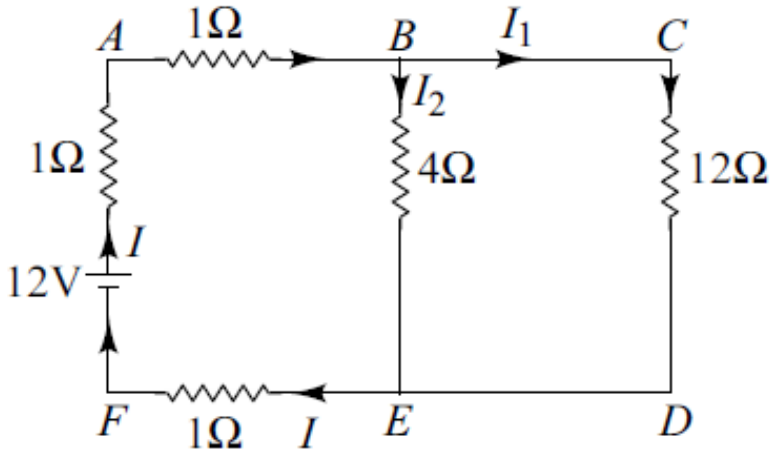
Applying the loop rule to loop $BCDEB$**,**

$$12I_1 - 4I_2 = 0$$
$$\Rightarrow I_2 = 3I_1 \text{ (2)}$$

Using (2) in (1), we get $I_1 = 0.5$ A

$$\therefore V_B - V_D = 12I_1 = 12 \times 0.5 = 6 \text{ V}$$

So the correct choice is (c).



Q.4 Answer:

1.7 A

Solution:

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Applying the junction rule at P, the current in branch $PQ = 2 + 2 = 4$ A.
Using the junction rule at Q, the current in branch $QR = 4$ A $- 1$ A $= 3$ A
Using the junction rule at R, we get
$I + 1.3$ A $= 3$ A $\Rightarrow I = 1.7$ A

Q.5 Answer:

$\mathrm{m} = 12, \mathrm{n} = 2$

Solution:

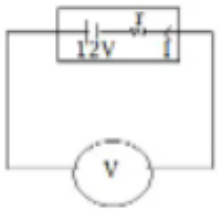
Let, we connect 24 cells in n rows of m cells, then if I is the current in external circuit then
$I = \frac{mE}{mr/n + R}$
For I to be maximum, $(\mathrm{mr} + \mathrm{nR})$ should be minimum.
It is minimum for $\mathrm{R} = \frac{\mathrm{mr}}{\mathrm{n}}$
So maximum current in external circuit is
$\mathrm{I} = \frac{\mathrm{mE}}{2\mathrm{R}}$
here $\mathrm{R} = 3, \mathrm{r} = 0.5$ so equation (2) become $\frac{\mathrm{m}}{\mathrm{n}} = 6$
so $\mathrm{n} = 2,\ \mathrm{m} = 12$

Q.6 Answer:

15 V

Solution:

Here the terminal potential, $\mathrm{V} = 12 + \mathrm{Ir} = 12 + \left(60 \times 5 \times 10^{-1}\right) = 12 + 3 = 15$ V



Q.7 Answer:

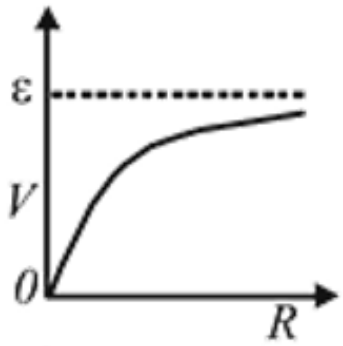
$\varepsilon, r/n$

Solution:

$\frac{1}{Net_r} = \frac{1}{r} + \frac{1}{r} \ldots$ ntimes
$\frac{1}{r_{\text{net}}} = \frac{n}{r}$
$r_{\text{net}} = \frac{v}{n}$
and emf $= \epsilon$

Q.8 Answer:

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Solution:

Current in the circuit,

$I = \frac{\varepsilon}{R+r}$

Potential difference across R. $V = IR = \left(\frac{\varepsilon}{R+r}\right) R = \frac{\varepsilon}{1+\frac{r}{R}}$

When $R = 0, V = 0\ R = \infty, V = \varepsilon$

Q.9 Answer:

0.5Ω

Solution:

Given : emf $\varepsilon = 2.1$ V

$\mathrm{I} = 0.2$ A, $\mathrm{R} = 10\Omega$

Internal resistance $\mathrm{r} =?$ **From formula.** $\varepsilon - \mathrm{Ir} = \mathrm{V} = \mathrm{IR}$

$2.1 - 0.2\mathrm{r} = 0.2 \times 10$

$2.1 - 0.2\mathrm{r} = 2$ **or** $0.2\mathrm{r} = 0.$

$\Rightarrow \mathrm{r} = \frac{0.1}{02} = 0.5\Omega$

Q.10 Answer:

$0.5\mathrm{ohm}$

Solution:

$\mathrm{r} = \mathrm{E}/\mathrm{I} = 1.5/3 = 0.5\mathrm{ohm}.$

Q.11 Answer:

$(n\varepsilon - 6\varepsilon), nr$

Solution:

Since due to wrong connection of each cell the total emf reduced to 2ε **then for wrong connection of three cells the total emf will reduced to** $(n\varepsilon - 6\varepsilon)$ **whereas the total or equivalent resistance of cell combination will be** nr

Q.12 Answer:

8.5 V

Solution:

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Terminal voltage

$= \varepsilon - IR$

$= 10 - 3 \times .5$

$= 10 - 1.5 = 8.5$ V

Q.13 Answer:

1

Solution:

Given : Number of cells, $n = 5$**, emf of each cell** $= E$ **Internal resistance of each cell** $= r$ **In series, current through resistance** $R\ I = \frac{nE}{nr+R} = \frac{5E}{5r+R}$
In parallel, current through resistance $R\ I' = \frac{E}{\frac{r}{n}+R} = \frac{nE}{r+nR} = \frac{5E}{r+5R}$
According to question, $I = I' \therefore \frac{5E}{5r+5R} = \frac{5E}{r+5R} \Rightarrow 5r + R = r + 5R$
or $R = r \quad \therefore \frac{R}{r} = 1$

Q.14 Answer:

zero

Solution:

The given circuit is

Equivalent emf is given by $\varepsilon_{eq} = \frac{2\times1}{2+1} = \frac{2}{3}$
we have,

$\frac{E_{eq}}{\varepsilon_{eq}} = \frac{E_1}{\varepsilon_1} - \frac{E_2}{\varepsilon_2}$

$\frac{\varepsilon_{eq}}{\left(\frac{2}{3}\right)} = \frac{2}{2} - \frac{1}{1}$

$\Rightarrow E_{eq} = 0\text{volt}$

Q.15 Answer:

$\frac{1}{3}A$ and 3 V

Solution:

$$I = \frac{8-4}{1+2+9} = \frac{4}{12} = \frac{1}{3}A$$

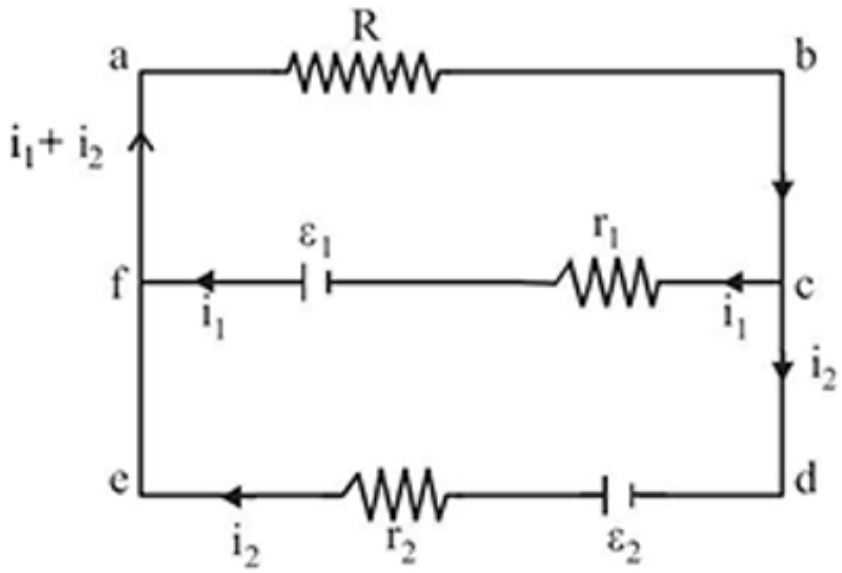
$$V_P - V_Q = 4 - \frac{1}{3} \times 3 = 3 \text{ volt}$$

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Q.16 Answer:

$\varepsilon_1 - (i_1 + i_2)\,R - i_1 r_1 = 0$

Solution:



Applying Kirchhoff 's rule in loop abcfa

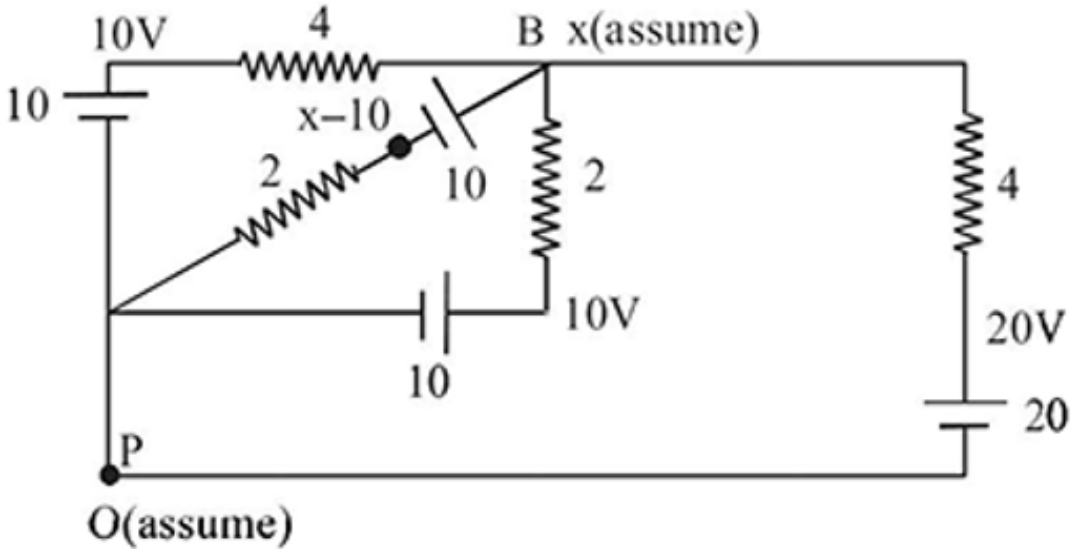
$\varepsilon_1 - (i_1 + i_2)\,R - i_1 r_1 = 0$

Q.17 Answer:

$\frac{25}{12}$ A

Solution:

The simplified circuit is



We have to find I. Let potential of point P be 0. Potential at other points are shown in the figure apply Kirchoff's current law at B where potential is assume to be x volt. $\frac{x-10}{4} + \frac{x-10}{2} + \frac{x-20}{4} + \frac{(x-10)-0}{2} = 0$

$\Rightarrow x - 10 + 2x - 20 + x - 20 + 2x - 20 = 0$

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$\Rightarrow 6x = 70 \Rightarrow x = \frac{35}{3}$ **volt**

$\therefore \quad I = \frac{20 - \frac{35}{3}}{4} = \frac{25}{12}$ A

Q.18 Answer:

conservation of energy

Solution:

Kirchhoff's first law is based on conservation of charge and Kirchhoff's second law is based on conservation of energy.

Q.19 Answer:

3 A from A to B

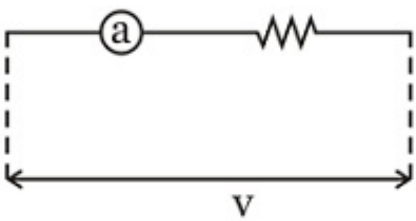
Solution:

Kirchhoff's junction rule states that the algebraic sum of all currents into and out of any branch point is zero : $\Sigma I = 0$**. By convention, the sign of current entering a junction is positive and current leaving a junction is negative.** $4\ A + 5\ A - 6\ A + I_{AB} = 0$**, therefore** $I_{AB} = 3\ A$**. The wire between points** A **and** B **carries a current of** 3 A **away from the junction.**

Q.20 Answer:

100

Solution:



$V_{max} = I_{max}R_g + R_s$

$= 10 \times 10^{-3}1000$

$= 10$ V

L.C. $= 0.1$ V

NL.C. $= V_{max}$

$\Rightarrow \quad N = \frac{10}{0.1} = 100$

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